KATTA NAVEEN

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"Gotta listen; whether it's humans or stars, everyone has a story",
Interested in exploring fundamental physics through compact objects, observational studies of compact objects, radio astronomy instrumentation, and computational astrophysics.

Education

2020–25 **Bachelor of Science - Master of Science dual degree program (BS-MS) in Physics**, Indian Institute of Science Education and Research, Bhopal.

2018–20 **Senior Secondary Education**, Presidency Junior College Andhra Pradesh State Board.

Research Experience

Sept. 2025–Present **Visiting Student**, Radio Astronomy Lab (RAL), Raman Research Institute (RRI), India. *Supervisor: Dr. Prabu T*

- Using 300+ days of single-pulse observations at 175 MHz (focused on PSR J1921+2153) to study magnetospheric processes, subpulse drifting, and phase-locked intermitten brightness (PLIBs).
- Conducting targeted low-frequency observations and analysis of the Galactic magnetar SGR J1935+2154, motivated by its association with FRB 200428.
- Developed an FRB-processing pipeline for DART, automating radio time-series processing, ADC-to-filterbank conversion, and parallel computation using multi-processing.
- Worked on the digital receiver backend and continuing on GPU acceleration for RFI mitigation and real-time transient detection.

May 2024–July 2025 **Master's Thesis** *link* , Compat Objects ASTrophysics (COAST Group), IISER Bhopal, India. *Supervisor : Dr. Ritam Mallick*

- Investigated the existence of quark matter and hadronic-to-quark phase transitions in neutron star interiors to model the *pasta phase*
- Developed a polytropic Equation of State (EoS) framework to model phase transition smoothness (first- to second-order), facilitating model-independent analysis consistent with astrophysical constraints.
- Analysed phase transition strength impacts on key observables, including mass-radius relations, compactness, tidal deformability, and twin-star configurations.
- Currently extending the model to evaluate neutron star surface tension using alternative EoS formulations.

May 2023–July 2023 **Research Intern**, Gauribidanur Radio Observatory, Raman Research Institute, Bangalore, *Supervisor : Dr. Prabu T*

- Development and commissioning of a two-element log-periodic dipole antenna interferometer for the continuous study of solar, galactic activity and also developed a remote acquisition system.
- Observed and analysed data of the sun and galaxy for abt one month, it is observing solar burts of all types for about three years and played a key role in solar weather and monitoring at low frequency observations.

Publications

2026 **Time-Domain Astronomy using Diamond Array Radio Telescope (DART)**. *IEEE Xplore; Accepted August 2026*

2025 **Innovative Web Tool for Remote Data Acquisition and Analysis: Customized for SKA Low Frequency Beamforming Test Bed LPDA Array at Gauribidanur Radio Observatory**. *Journal of Astronomy and Astrophysics; Published 2025*, *link*.

Projects

Dec. 2022–Apr. 2023 **Pulsar Timing & Signal Processing**, IISER Bhopal, India
Studied pulsar periodicity, dispersion and detection significance, with practical pulsar detection and data analysis using PRESTO.

Aug. 2024–Jan. 2025 **Machine Learning for Strong Gravitational Lens Detection** , IISER Bhopal, India
Developed a convolutional neural network to identify strong gravitational-lens candidates, worked with image preprocessing, supervised classification, model training, and evaluation.

Outreach

Feb 2026 **Radio Astronomy Winter School (RAWS) 2026**. *Gauribidanur Radio Observatory, Raman Research Institute*.

- Gave tutorials on the end-to-end process of pulsar detection, from searching time-series data for periodic signals to identifying pulsar candidates, including an introduction to dedispersion and its role in pulsar searches.
- Delivered an introductory talk on neutron-star emission processes, introducing the physical mechanisms underlying radio emission from neutron stars.
- Also was a part of the team giving hands-on interferometry for solar observations and H1 line detections with horn antenna.

Conference Proceedings

2026 **Web Based Pulsar Observation and Analysis**,
In-house symposium Raman Research Institute 2026, Poster

2025 **Investigating Max Pressure Jump of mixed phase in Neutron Stars**,
IISER, bhopal Physics Symposium Poster

2024 **Two Element Broadband Log-Periodic Dipole Antenna Radio Interferometry: Exploring the SKA Low Frequency Band**, 42nd Meeting of the Astronomical Society of India (ASI–2024), Poster

2023 **Innovative Remote Web Tool for SKA Test LPDA Array at Gauribidanur Radio Observatory**, 4th Annual Meet of Modern Engineering Trends in Astronomy (META–2023), Poster

Awards & Honors

2025 Qualified GATE, Paper – Data Analysis 2025, 2026; Physics 2026
2016–20 National Means-cum-Merit Scholarship, Andhra Pradesh
2017 INSPIRE Award for Science Projects, Andhra Pradesh

Attended Workshops

2026 **HSF India Gravitational Wave Workshop,** Inter-University Centre for Astronomy and Astrophysics (IUCAA).

2025 **AI/ML Applications to Astronomy and Astrophysics**, Inter-University Centre for Astronomy and Astrophysics (IUCAA).

2024 **(Sub)Millimeter Astronomy: Introduction to ALMA and Indian Perspectives**, 42nd Meeting of the Astronomical Society of India.

2024 **Sagan Summer Workshop**, NASA Exoplanet Science Institute; hands-on sessions, expert talks, and discussions on exoplanet science.

References

Dr. Prabu T Raman Research Institute, Bengaluru
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Dr. Ritam Mallick Indian Institute of Science Education and Research Bhopal
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